

Cooperative Heavy Lifting Using Unmanned Multi-Agent Systems

Yu Heng Tan, Shupeng Lai, Kangli Wang and Ben M. Chen

Department of Electrical & Computer Engineering

National University of Singapore

Singapore

Abstract—Cooperative load carrying is an area of multi-agent systems that has extensive potential applications. Though most of the complexities involved with multi-agent operations have been investigated in general swarm operations, the greatest challenge of heavy duty aerial transportation that has yet to be addressed is the limited control authority of the system as a whole as each individual agent, and hence effector, is typically significantly smaller than the payload. Based on the configuration whereby the vehicles are connected rigidly to the load, we present a general optimisation method to find the most effective team arrangement that will maximise the control authority within the physical constraints of the system. Additionally, each UAV is allowed to tilt laterally at a fixed angle relative to the payload, which provides a larger maximum rotational acceleration achievable about the yaw axis. The optimised solution, implementation and results based on the cooperative carrying task in the 2017 International Micro Air Vehicles Competition is presented in this paper. Testing results show that the system designed is effective in transporting the payload autonomously according to generated trajectories.

I. INTRODUCTION

Aerial transportation is an emerging industry which has seen increasing development and involvement of industrial players including Amazon [1] and DHL [2] with the aim of automating consumer deliveries. However, one of the major challenges of doing so is being limited by the payload restrictions of the vehicle used. Although the maximum thrust of the vehicle determines the absolute maximum payload that it can carry, transporting excess weight also alters the performance and flight times of the system. Intuitively, using larger vehicles will allow for a larger maximum payload, though there is a limit to the degree which this can be extended to especially for aerial vehicles. This is because the payload-to-vehicle weight ratio scales poorly for UAVs as the higher weight of a larger propulsion system needed to carry more payload simultaneously adds to the lifting power requirements.

Cooperative carrying using a team of smaller agents is an alternative concept to overcome this challenge. In addition, smaller agents have the advantage of being both less mechanically complex and more affordable. A cooperative team can also be flexibly rearranged according to the needs of different tasks, which will be cost efficient feature that is useful in industrial applications. The complexities are, however, the need to account for the communication and overall control



Fig. 1. The unmanned system developed, which consists of five multi-rotors, lifting and transporting a square frame payload autonomously.

of the full system which becomes increasingly complex as the number of agents involved increases.

The basic concept of inter-agent cooperation needed have been demonstrated with effectively coordinating the different agents in a swarm [3]. Path planning algorithms for such multi-agent systems have also been presented for specific tasks such as basic collision avoidance [4] and using the combined knowledge of the team to cooperatively follow a target [5]. However, there has been very little documented use of such multi-agent systems to transport large and heavy loads. Carrying large and heavy loads presents a new challenge in addition to the coordination needed for multi-agent systems as the size of each agent is significantly smaller than the load to be lifted. Although the total lifting power needed can be achieved through the linear combination of thrust provided by every agent, accurate position control of the full system is much more challenging as the control authority is diluted by virtue of the force inputs being distributed.

There are countless possible ways to physically implement cooperative carrying. Specifically, the variable parameters in any cooperative carrying system includes the number of vehicles in the team, the structure of the system and the control method used. As such, a system design approach can be used to resolve the challenges above by selecting suitable options to create the most effective configuration. As the variables are interrelated through multiple complex relationships, the systems based on different configurations could result in changes

in performance that are not directly intuitive. Therefore the general task of cooperative carrying can be framed as an optimisation problem that aims to obtain the optimal system configuration that will maximise performance.

In this paper, we first review the previous work done in the usage of aerial vehicles for cooperative transportation in Section II. Section III then presents a general optimisation method proposed to find the best configuration for any given number of agents and a known flat payload. A specific solution based on the cooperative carrying task in the 2017 International Micro Air Vehicle (IMAV) competition was then derived, and the physical implementation is subsequently shown in Section IV with the model of the system is presented in Section V. Lastly, the performance of the system designed and the results from IMAV tasks are analysed in Section VI and conclusions from the work presented are drawn in Section VII.

II. RELATED WORK

The majority of previous work in this area assumes that flexible solid material is used to join the agents and the payload. Though in reality the connections can deform in shape, they can be treated as unstretchable rigid objects if they are tightly stretched, which is a valid assumption as long as the vehicle is not accelerating downwards. The swing of a payload carried this way beneath the vehicle can be countered accordingly through a suitable trajectory found through dynamic programming as proposed in [6]. [7], [8] presents a full model of such an underslung system and proposes a team consisting of three UAVs, which form a system that ensures static equilibrium. Furthermore, cooperative carrying with additional trajectory planning and obstacle avoidance was implemented in [9], which presents a general framework to deal with multi-agent systems. Alternatively, the links between the UAVs and the payload can be more realistically modelled as stretchable and not just deformable as presented in [10]. The additional variable of the cable tensions in this case were treated as a disturbances to the system which the control system should eliminate, which was demonstrated using non-linear techniques.

Unlike the nature of the systems above, the payload can also be rigidly mounted to the UAVs, as proposed by [11]. This implies that the carriers and payload effectively become a single vehicle. This greatly simplifies the model of the system, though the scale of a large rigid body implies that the moment of inertia of the system is much larger than each individual agent. Still, this method can be made to be stabilisable and implementable, and characteristically better suited to tasks which require heavy lifting than the flexible link solution as the larger disturbance force that the system needs to overcome in that approach might be insurmountable by the agents involved.

Previously, the concept of cooperative carrying has been widely modelled but seldom discussed as a system design problem. In addition, there is little evaluation on the effect of formation arrangement of a multi-agent team in such a problem. This is in fact, fundamental to effectively solving

the challenges in control and positioning accuracy, which is addressed in this paper below.

III. OPTIMISATION METHOD

For a team of multi-agents to cooperatively transport a relatively large payload, the configuration of the system is integral to the effectiveness of the solution. The optimal configuration is also highly dependent on the shape and nature of the payload. Here, we will make two assumptions to the problem while maintaining a general approach: firstly, the payload is assumed to be a 2 dimensional flat shape with a constant thickness; and secondly, the UAVs are attached to the payload rigidly. Considering the payload as flat shape is sufficient to represent the issues with control authority as multirotors are inherently able to generate greater acceleration in roll and pitch than yaw, and a flat object will have the largest rotational inertia in the yaw axis. Although systems with the payload connected via flexible links have been widely studied, the other possible connection method of rigidly mounting the vehicles to the frame creates an overall system with a dynamic model that is much more straightforward. The string-based approach results in a system with constantly changing dynamics; the nature of the connections between the UAVs and the payload results in the relative position of each unit to be inconsistent. On the contrary, a rigidly mounted system moves as a single body with less variables as the relative position of each UAV in the system is fixed, hence making the system significantly easier to model. Nevertheless, the overall system still needs to be adequately controlled for the solution to be valid. This method of payload attachment is chosen here as it allows greater flexibility for extending such a system to involve more agents, unlike the method of aerial towing which becomes significantly more complex with larger teams.

The greatest challenge of controlling a single large rigid body is the fact that the moment of inertia of the full system will naturally be much larger than the rotational acceleration that can be provided by the individual agents, particularly in the yaw axis for a flat object. The issue is further compounded if the point loads of each UAV are placed further away from the centre of the frame. In order to control the yaw rotation of the system effectively, the system has to be able to generate a rotational torque that is sufficiently large. The standard method used to control yaw in a multirotor by changing the rotational velocity of each counter-rotating set of propulsors is relatively ineffective in this case as each motor is small. Even with a large number of motors, the difference in rotational speeds of the counter-rotation still needs to be fairly large, which will tend to result in a loss of vertical thrust and hence lifting power when the rotational output of the motors are saturated. This significantly affects the system's ability to hold altitude.

To address this issue, we propose an alternative solution to increase the control authority of the system in the yaw axis by tilting the corner UAVs towards the mid-line of the system. This tilt angle redirects a component of thrust in the direction which will create a turning moment about the

yaw axis, hence the overall rotational acceleration that can be achieved by the system with all agents contributing a moment arm will then be much more than the achievable torque of the conventional method. The amount of torque that can be produced increases as the distance of the vehicle from the centroid of the payload increases. However, this will simultaneously increase the moment of inertia of the system in the yaw direction as previously established. This then becomes an optimisation problem to find the most suitable tilt value and arrangement of each UAV in the system so as to improve the control authority without relying on an overly large tilt angle that will compromise the vertical lifting ability of system.

A. Problem formulation

Here, a general formulation of the problem is presented so that the optimisation of the objective function corresponds to the optimal placement and pose of the multi-agent team with respect to the payload while ensuring that the system can maintain static equilibrium. We will first fix the number of agents in the UAV team to simplify the problem. This can be intuitively decided based on the shape of the payload, or alternatively treated as an additional variable to be iterated accordingly to find the optimal value. In addition, we also set one UAV to be positioned at the centroid of the payload shape. This placement allows the central UAV to effectively estimate the states of the full system, which can be subsequently used in the centralised control of the team. This positional placement would also imply that this vehicle has little effect on the rotational acceleration of the system about all axes, therefore the primary function of this UAV is to contribute to total lifting power and hence will not be included in the calculations associated with rotational control authority below.

The adjustable system parameters in the problem are: the placement points of the agents on the payload frame and the value and direction of tilt of each individual vehicle. Therefore, for a team made up of n agents, the total number of optimisation variables would be $2n$. The 2-dimensional placement point of each UAV is treated as one variable, represented by integers, with each integer representing a discretised position on the frame.

To model the control authority in all three rotational axes and the vertical translation direction, the objective function to be maximised is shown in Equation (1).

$$z = k_R \frac{\tau_y}{I_{yy}} + k_P \frac{\tau_x}{I_{xx}} + k_Y \frac{\tau_z}{I_{zz}} + k_T \frac{F_z}{m} \quad (1)$$

This is formulated by linearly combining the weighted sum of the rotational acceleration about all three axes (roll, pitch and yaw) and the linear acceleration in the lifting direction. For the three rotational axes, this is given by the torque about each of the axes, τ_x, τ_y, τ_z , divided by the corresponding rotational inertia I_{xx}, I_{yy}, I_{zz} . For the acceleration in the lifting direction, the control authority is found from the total vertical thrust $F_z = \sum_{i=1}^n T_{zi}$, where T_{zi} is the component of thrust from each agent that is acting in the lifting direction,

divided by the total mass m . Scaling factors k_R, k_P, k_Y, k_T are incorporated into the formulation to allow us to balance the relative cost of each value.

The constraints of the problem are then defined by the physical limitations of the system and ensuring that static equilibrium is guaranteed. These are expressed as

$$\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \leq 1.1d \quad \text{for all } 1 \leq i, j \leq n, i \neq j \quad (2)$$

$$\sum_i x_i = 0 \quad 1 \leq i \leq n \quad (3)$$

$$\sum_i y_i = 0 \quad 1 \leq i \leq n \quad (4)$$

$$\sum_i y_i \sin \beta_i = 0 \quad 1 \leq i \leq n \quad (5)$$

$$\frac{1}{n} \sum_i \cos \beta_i > \tau \quad 1 \leq i \leq n \quad (6)$$

where x, y are the 2D positional coordinates on a plane where the centroid of the payload is the origin, d is the tip-to-tip diameter of each UAV, β is the tilt angle, τ is the minimum thrust factor, and n is the total number of UAVs in the team.

Equation (2) ensures that the UAVs are adequately spaced apart so that there is no physical overlapping of the vehicles. A minimum safety distance of $1.1d$ is enforced between the agents. The net torque and force of all agents in the team should be zero at a hovering state to ensure static equilibrium. Assuming that all the agents involved are identical, which is a fair assumption that is normally true in multi-agent teams, this requirement is formulated in Equation (3) and (4). As the system is defined with its centre of mass at the origin, this requirement is equivalent to enforcing the average position of all agents along both roll and pitch axes to be zero. The case where there is a non-zero net torque can be equilibrated through applying counter-acting moments through individual agents. However, this would naturally be a suboptimal solution and therefore not discussed as a possible solution. Similarly for the yaw axis, the net torque created by the tilt angle is enforced to be zero (Equation (5)). Increasing the tilt angle will increase the achievable rotational acceleration in yaw. However, this comes at the price of sacrificing lifting thrust. Therefore, there has to be a limit for the tilt angle such that the system is still able to generate sufficient vertical lift. This is written as the constraint in Equation (6) with the optimisation parameter τ . This value is found from the thrust force of each agent and the all up weight of the full system. Lastly, we also impose a range for the tilt angle $-\frac{\pi}{2} < \beta < \frac{\pi}{2}$ which intuitively ensures that the vehicle is mounted on the top side of the payload.

B. Derived solution

The implementation of the method above was performed based on the cooperative carrying task in the 2017 International Micro Air Vehicle (IMAV) competition in Toulouse, France, where a team of UAVs are required to autonomously transport a payload frame a fixed distance to a specified target

position. The actual payload is a square frame made from wood with a crossbeam through the middle of each planar axis which weighs 2 kg with dimensions 1×1 m.

An evolutionary algorithm (EA) was used to find the optimal solution for this task using the constraints and objectives defined in Section III. EA is suitable for complex problems that are not directly intuitive like the one presented here as it only requires the objective function and constraints to be well formulated. The solution is then found iteratively through each subsequent generation until the predefined convergence conditions are met [12]. For a standard square-shaped payload, four agents would be the theoretical minimum to obtain static equilibrium while simultaneously producing a system with available actuators in all channels. Hence, four vehicles will be used in the optimisation below. The UAVs used have a diameter of 450 mm, weigh 0.55 kg and have a maximum thrust of 1.2 kg. The optimisation parameters selected are shown in Table I.

TABLE I
PARAMETERS USED IN THE OPTIMISATION PROCESS.

k_R	10
k_P	10
k_Y	200
k_T	1

The results found from the optimisation process are shown in Figure 2, with the circles denoting the diameter of each UAV and the arrows identifying the tilt direction and relative magnitude of β . The solution found places an agent at each corner of the frame with a tilt of 11.4 degrees towards the central line of symmetry of the system. This arrangement is identical to the configuration of a conventional quadrotor, if each UAV is seen as a ‘propeller’ in this layout. As such, controlling the UAVs in this multi-agent system can be simplified into the equivalent of controlling a single quadrotor.

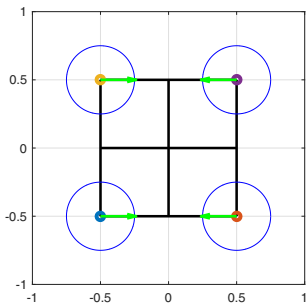


Fig. 2. The optimisation process identifies the optimal configuration of placing the four UAVs at each corner with an inward tilt.

IV. IMPLEMENTATION

Centralised control was used to coordinate the thrust outputs of each UAV, with the central agent being the controller of the full system, which is suitable as its pose estimation sensors are

at the centre of the system. The full assembled system with all agents attached to the payload is shown in Figure 3, and the overall parameters of the full system are shown in Table II.

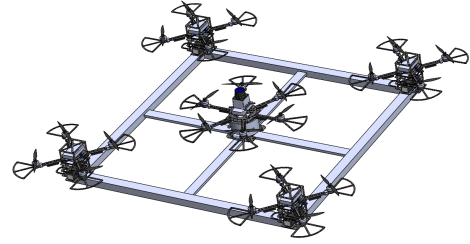


Fig. 3. A visualisation of the UAV team in the optimised positions for the square frame payload.

TABLE II
PHYSICAL PARAMETERS OF THE UAV TEAM WITH PAYLOAD.

Maximum diameter (mm)	1864
Total weight (kg)	5.5
I_{xx} (kg/m ²)	0.79068
I_{yy} (kg/m ²)	0.79169
I_{zz} (kg/m ²)	1.55874

A. Vehicle design

The UAVs used were designed and manufactured at the National University of Singapore. Here, two different types of vehicles, quadrotors and hexarotors, are used in the team. Four quadrotors take the place of the four agents that are attached to the corners of the frame and one hexarotor was used as the central UAV (Figure 3). Using a hexarotor in the central position increases the total lift generated by the system and hence reduces the amount of thrust that the corner agents need to produce. This allows us to minimise the size and weights of the corner agents, thereby reducing the point loads acting in the corners which increases the moment of inertia of the system. The quadrotors consist of carbon fibre arm structures with a core frame made of carbon fibre plates connected with aluminium stand-offs. The hexarotor was constructed similarly, with a larger frame to accommodate a six arm layout. The dimensions and components are summarised in Table III below. Each vehicle also consists of their individual flight controllers with a corresponding power distribution board that is typical of a standard multirotor. Additionally, the hexacopter is equipped with sensors for navigation and localisation depending on the area of operation as described in Section IV-D.

B. Communication links

As the system operates on a centralised control approach, each corner quadrotor receives its command signal directly from the central UAV. The changes in state of the system are measured directly by the central UAV, hence no relay of

TABLE III
PARAMETERS OF THE TWO TYPES OF UAV USED.

Parameter	Type 1 (Quadrotor)	Type 2 (Hexarotor)
No. of motors	4	6
Length (tip to tip) (mm)	420	600
Weight (kg)	0.55	1.1

information from the corner quadrotors back to the main controller is needed. As such, a direct wire connection provides the fastest and most reliable solution and is adopted here. The quadrotors can either operate individually or receive flight signals from the main controller when working as part of a cooperative team. A switch plug design allows the corner UAVs to switch easily between operating as part of the team to operating individually.

C. Mounting bracket

To secure the UAVs to the payload, a customised bracket was designed and manufactured. The 3D printed corner bracket allows the robot to be aligned with the corners of the frame, and hence is a flexible attachment that can also be used with similar square or rectangular frames with different dimensions. The payload is secured to the UAVs using cable ties which thread through the holes in the mounting bracket. On top of the mounting bracket, a wedge angle is added to set the UAVs at the desired tilt angle found in Section III. Two wedge pieces are used to securely attach each mount and to the UAV, as shown in Figure 4.

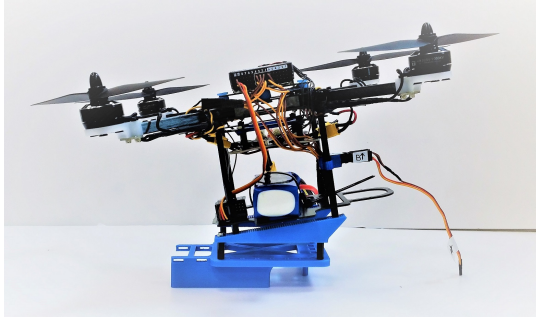


Fig. 4. The assembled mounting bracket and tilt piece with a corner quadrotor.

D. Positioning and localisation

The mission management of the system is controlled on the central hexacopter, which runs a custom ROS-based UAV operating system on a 4GB RAM UP Board that is carried onboard. As the corner UAVs are direct slaves receiving command signals from the central controller, they do not require any form of processing power beyond the basic flight stack which will allow it to operate both as part of the team and individually. Localisation and positioning is done using the input from various sensors onboard the central hexacopter.

When operating outdoors, a GPS receiver attached to the central hexarotor allows navigation to be performed when

provided with GPS coordinates of the target locations. In GPS-denied environments, 2-dimensional laser SLAM is used for localisation and navigation. The horizontal plane movements are measured using a 20 m Hokuyo laser scanner and a single beam laser pointed towards the ground provides the vertical measurement.

V. MODELLING & CONTROL

The control structure of any unmanned autonomous vehicle consists of the elements: mission management, path planning, trajectory generation, outer and inner loop control. The system here is effectively controlled as a single vehicle as the agents are rigidly mounted and directly taking signals from the centralised controller. The components such as mission management, path planning and trajectory generation which are higher up in the control architecture are based on earlier work done by our team [13], [14]. As previously identified, the overall structure of our system and hence the position and attitude control of the system is close to that of a regular quadrotor, with minor changes applied to in the way the yaw channel is controlled.

The global position of a body can be defined in the inertial frame position $\mathbf{P}_n = [x, y, z]^T$, with the angular position defined by the Euler angles $\Theta = [\phi, \theta, \psi]^T$. In the body frame, the velocity and angular velocity are expressed as $\mathbf{V}_b = [u, v, w]^T$ and $\omega_b = [p, q, r]^T$ respectively, with the rotational and translational motions associating the two coordinate systems expressed as the following well established navigation equations [15]:

$$\dot{\mathbf{P}}_n = \mathbf{R}_{n/b} \mathbf{V}_b \quad (7)$$

$$\dot{\Theta} = \mathbf{S}^{-1} \omega_b \quad (8)$$

where the rotational matrix $\mathbf{R}_{n/b}$ and lumped transformation matrix $\mathbf{S}^{-1}$ are

$$\mathbf{R}_{n/b} = \begin{bmatrix} c_\phi c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix} \quad (9)$$

$$\mathbf{S}^{-1} = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi / c_\theta & c_\phi / c_\theta \end{bmatrix} \quad (10)$$

and $s_* = \sin(*)$, $c_* = \cos(*)$ and $t_* = \tan(*)$.

The rigid body dynamics, as in the case of a standard quadrotor, can then be expressed as

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \frac{1}{m} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad (11)$$

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \mathbf{I}^{-1} \left(\begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times \mathbf{I} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \right) \quad (12)$$

The force and moment of the system can be expressed as

$$\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ |T_{z1}| + |T_{z2}| + |T_{z3}| + |T_{z4}| + |T_{z5}| \end{bmatrix} + mg \begin{bmatrix} -\sin \theta \\ \sin \phi \cos \theta \\ \cos \phi \cos \theta \end{bmatrix} \quad (13)$$

$$\begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} \frac{l}{\sqrt{2}} (|T_{z2}| + |T_{z3}| - |T_{z1}| - |T_{z4}|) \\ \frac{l}{\sqrt{2}} (|T_{z1}| + |T_{z3}| - |T_{z2}| - |T_{z4}|) \\ \sqrt{2} (T_{y1} - T_{y3}) \end{bmatrix} \quad (14)$$

The subscripts 1 to 4 refers to the four corner UAVs, while subscript 5 is the central UAV. $T_{zi} = T_i \cos \beta_i$ is the thrust component that contributes to vertical lift, $T_{yi} = T_i \sin \beta_i$ is the thrust component that contributes to yaw control and l is the distance of the frame corner to the centre. At hover state, each individual agent is in equilibrium i.e. there is zero net torque, therefore there is no need for diagonally opposite vehicles to counter the torque of each other, unlike the setup of a conventional quadrotor. Based on the model above and the parameters of the system, the inner loop controller for the system was designed using a linear quadratic regulator (LQR), and we implement robust and perfect tracking (RPT) control proposed in [16] for the outer loop controller. The detailed derivation of such a controller for a similar system was presented in [17].

VI. EXPERIMENTS & RESULTS

The system was tested to operate both indoors and outdoors at the IMAV competition to demonstrate the performance of the system under both GPS and GPS-denied situations as well as other conditions such as wind disturbances. The system successfully navigated to and landed precisely on the given targets, transporting the payload 10 m in the indoor execution and 50 m outdoors. This performance achieved first place in both competitions and the demonstration of the system in action can be found at [18].

The experiments showed that even with strong wind conditions, the system was able to track the given trajectory closely. To complete the outdoor mission, the multi-agent system moved the payload a horizontal distance of 50 m while flying at a height of 5 m and velocity of 1 m/s. For the competition, the team first travels 5 m beyond the target to ensure mission completion before moving back for landing. The measured flight trajectory and reference is shown in Figure 6, and the corresponding tracking error is shown in Figure 7. Figure 6 shows that the performance in the tracking of the given trajectory is very good, with the team maintaining a maximum deviation 0.7 m of the reference at all times. The performance shows the effectiveness of our developed system and verifies the improved control authority of the configuration adopted.

For the indoor mission under a GPS-denied environment, the same system flew a distance of 10 m at a height of 1 m within the confined competition area. The 2D laser SLAM used for

positioning measurement gives localisation information with higher precision compared to GPS, allowing the system to navigate tight areas as there is less drift from positioning error. As with the case in the outdoor mission, the system first flew beyond the landing point before returning to land on the specified target as shown in the trajectory plot in Figure 8. The tracking error shown in Figure 9 shows the ability of the system to follow the trajectory precisely, with tracking error of less than 15 cm for most of the mission once the team is moving forwards. The maximum deviation happens during the transition between takeoff and forward flight, resulting in a relatively larger tracking error as the system starts moving forwards before it has reached the specified height. This performance also demonstrates that the positional accuracy of the system is highly correlated to the accuracy of the localisation method used, and that a large multi-agent system is responsive to the control method that is primarily based on a conventional quadrotor with the proposed design.

The competition demonstrations shown here successfully verifies the trajectory tracking ability of the system and overall effectiveness of the proposed solution. Furthermore, this configuration with the specified payload can achieve a flight time of around three minutes, which is equivalent to a horizontal distance of 150 m if flying at a speed of 1 m/s. This is relatively large, considering that the size of the individual agents used and individual lifting power is significant smaller than the weight and size of the payload frame.

VII. CONCLUSION

There is an infinitely large number of possible ways to execute aerial transportation using a team of UAVs. In order to find the optimal configuration, we apply an optimisation using EA to find the best position and pose placement of the UAVs on a specific payload that is flat in nature, using the rigid mount concept of payload attachment. The implementation based on the solution found provides a proof-of-concept of the ideas proposed that can allow for greater cooperative heavy lifting using multiple UAVs. Most importantly, the limited control authority, especially in the yaw axis, can be effectively improved by adding a tilt angle to the quadrotors placed along the outer edges. Further improvements to the system can be made by using releasable clamps to attach the vehicles to the payload, so that less human intervention is required for the process of lifting the load. Communications between the central controller and the other UAVs will also benefit if wireless transmission is used, which will aid the transition between cooperative teamwork to individual operation. Nevertheless, the results presented show that the system design process is successful in producing a cooperative carrying system that can autonomously transport a given payload with minimal tracking error, overcoming the existing challenges of limited control authority. Furthermore, the optimisation process proposed can also be applied to teams with higher number of agents and other payloads beyond a simple square frame where the optimal solution might be less intuitive.



Fig. 5. The full system performing in the outdoor and indoor tests.

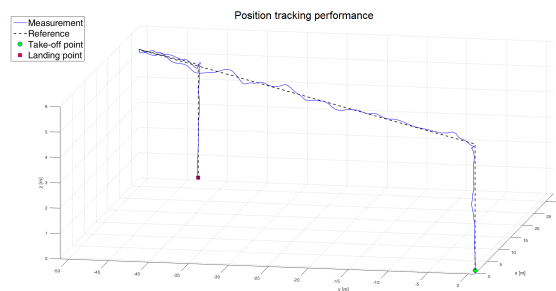


Fig. 6. The measured path of the system relative to the reference given for the outdoor mission.

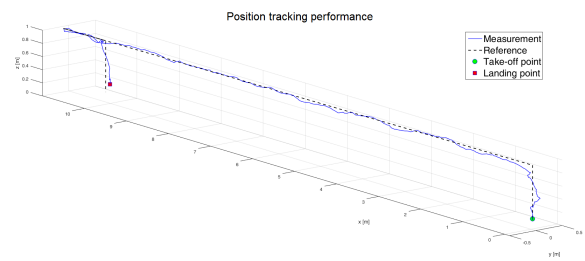


Fig. 8. The measured path of the system relative to the reference for the indoor mission.

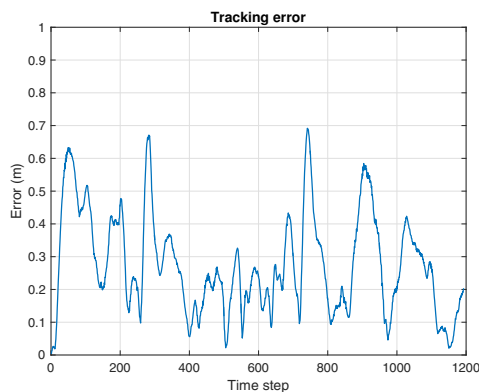


Fig. 7. The tracking error in the outdoor mission.

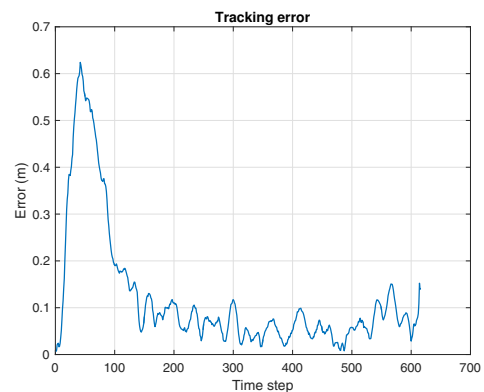


Fig. 9. The tracking error in the indoor mission.

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